

1 Quick revision of differentiation

Differentiate the following: (The first question of exam 1 is almost **always** differentiation)

- $x^3 + 2x - 1$
- $3x^2 \sin(2x)$
- $e^x \cos(x)$
- $3 \tan(x)$ (Do this using quotient rule)
- $\ln(x^2 + 1)$

2 Functions

- what's a mapping/relation? A fixed map of inputs to outputs from an input set to an output set (draw a quick diagram)
- What's a function? A mapping from one set to another that is one-to-___ (many/one)
- How do we write a function? $f : Dom \rightarrow \mathbb{R} : f(x) = \dots$
 - $\mathbb{R}$ here is NOT the range!! DO NOT mistake it for the range. (It is a bigger set that the range lives in, called a *codomain*)
- What's a Domain and Range?
 - Domain: Set of all **possible** inputs
 - Range: Set of all **actual** outputs
 - eg: $f : \mathbb{R} \rightarrow \mathbb{R} : f(x) = x^2$
- What's an implied domain? When you are given a function without a domain specified, the implied domain is the **largest** possible domain that makes the function work.
 - eg: $f(x) = \frac{1}{x}$
 - eg: $g(x) = \sqrt{x - 3}$

Some more definitions:

- one to one vs one to many functions:
- odd and even functions
- Continuity: A function that is continuous has no breaks, jumps or holes in it.

$$f : (-\infty, a] \rightarrow \mathbb{R}, f(x) = x^3 + 2x^2$$

For what values of a is f a one to one function?

2.1 Hybrid/piecewise functions

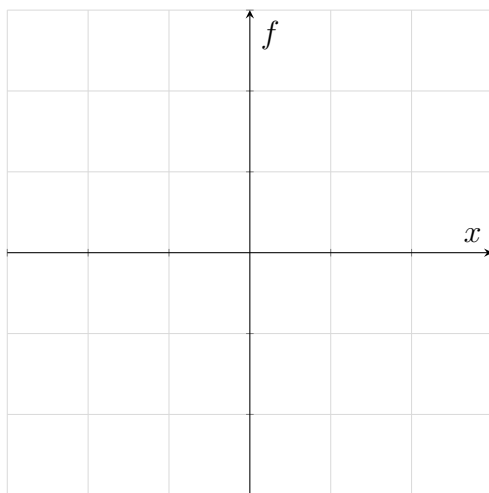
A function with different expressions over different parts of the domain eg:

1.

$$f(x) = \begin{cases} x^2, & x < 0 \\ 2x + 1, & x \geq 0 \end{cases}$$

Think of this as a function made up of two parts.

(a) Plot f



(b) Is f continuous everywhere?

2.

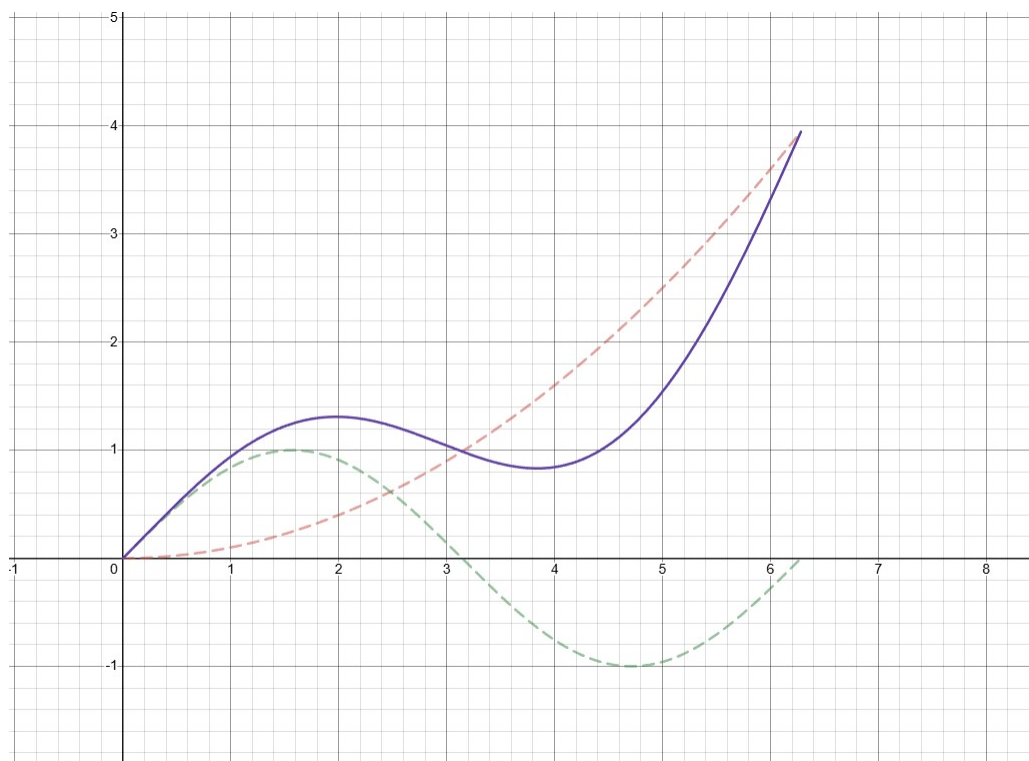
$$g(x) = \begin{cases} x^2 & x < a \\ e^x & \text{elsewhere} \end{cases}$$

(a) Are there any values of a that makes g continuous everywhere?

3 Addition of ordinates review

adding y -values for a given x -value

Consider graphing $f : (0, 2\pi) \rightarrow \mathbb{R}, f(x) = \frac{x^2}{10} + \sin(x)$



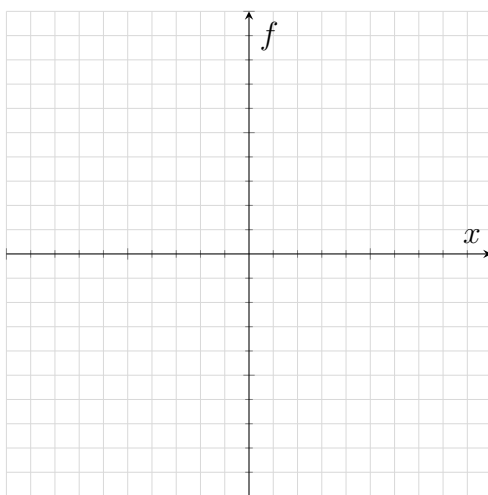
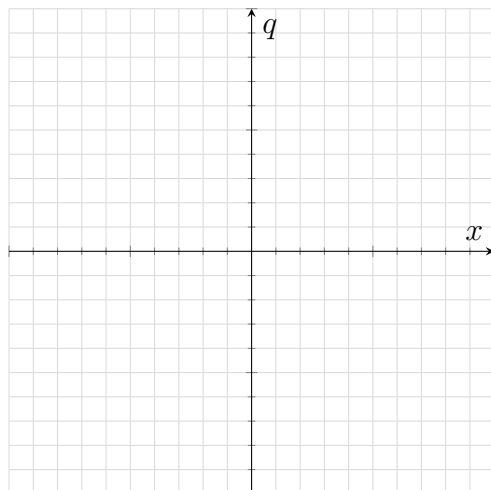
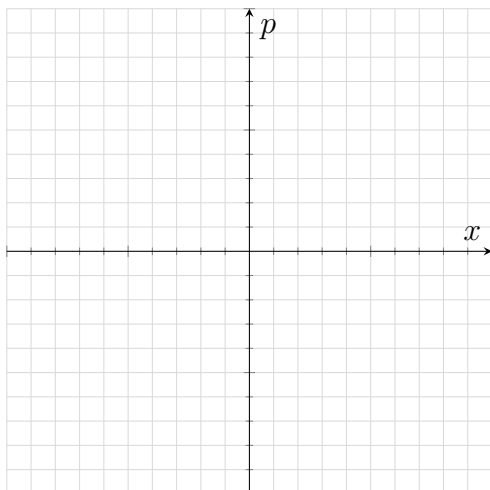
strategy?

- start with finding key points of each curve (if there are any)
- plot the curves individually
- add the height of each curve at discrete intervals (i.e at every tick)
- make sure the y values at key points match

New way to think of dilation from the y axis! i.e : $f(x) = 2x^2 = x^2 + x^2$

Sketch the following, and then double check with your calculator

- $p(x) = 2x + \sqrt{x}$
- $q(x) = e^{-x} + 3x$
- $f : (-2\pi, 2\pi) \rightarrow \mathbb{R}, f(x) = \sin(x) + \cos(2x)$



4 Composite functions

A function inside another function/

A function passed into another function/

A function inputted into another function/

A function of a function

Let $f(x) = x^2 + 2x$, and $g(x) = (x - 1)^2$

- Before: $f(x) = x^2 + 2x$

– x was inputted into f and evaluated according to the rule $x^2 + 2x$

– eg: to get $f(2)$: $f(2) = 2^2 + 2(2) = 4 + 4 = 8$

- $f(g(x))$ means

$$x \longrightarrow g(.) \longrightarrow f(.)$$

eg: To find $f(g(2))$

$$2 \longrightarrow g(2) = (2 - 1)^2 = 1$$

$$\longrightarrow f(1) = 1^2 + 2(1) = 3$$

$$\text{i.e } f(g(2)) = f(1) = 3$$

To simplify $f(g(x))$, we substitute $g(x)$ into f wherever we see an x :

$$\begin{aligned} f(g(x)) &= f((x - 1)^2) \\ &= ((x - 1)^2)^2 + 2((x - 1)^2) \\ &= (x - 1)^4 + 2(x - 1)^2 \end{aligned}$$

Try to simplify $g(f(x))$.

4.1 When composite functions can exist

Consider $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = 2x - 4$ and $g : \mathbb{R} \rightarrow \mathbb{R}, g(x) = \sqrt{x}$

- Simplify $f(g(x))$.
- Simplify $g(f(x))$.
- Find $f(g(1))$
- Find $g(f(1))$

Sometimes the output values from the inner function cannot be taken by the outer function. In those cases the composite function does not exist.

Then when does $g(f(x))$ exist?

- When $\text{Ran } f \subseteq \text{Dom } g$
- i.e The values that f outputs must be within the domain of g

1. $f(x) = \sqrt{x-2}, g(x) = 3x + 12$ Does $f \circ g$ exist over $\mathbb{R}$?

- $f \circ g$ ('f of g') is another way to write $f(g(x))$

2. We can restrict the domain of g so that $f \circ g$ exists. Find a suitable domain.

3. $f(x) = \ln x, g(x) = x^2 - 4$

- (a) Find a suitable domain for g s.t $f \circ g$ exists.
- (b) Find a suitable domain for f s.t $g \circ f$ exists.